

# WEEK 1: INTEGRATIVE PHYSIOLOGY

## RENAL PHYSIOLOGY

### FUNCTIONAL ANATOMY

#### THE NEPHRON

- Basic unit: each nephron = glomerulus + renal tubule

- Each human kidney has ~1 million nephrons

- Vary in structure species -> explains kidney size differences

- Glomerulus

- Tuft of capillaries (200  $\mu\text{m}$  wide) that invaginates into Bowman's capsule
- Blood supply:
  - Afferent arterial (larger diameter) -> brings in blood
  - Efferent arteriole (smaller diameter) -> carries blood out (drains)
- Filtration barrier:
  - Fenestrated capillary endothelium (70-90 nm)
  - Glomerular basement membrane (basal lamina)
  - Podocytes (specialized epithelial cells) with interdigitating pseudopodia -> form filtration slits (25 nm wide) covered by thin membranes
- Mesangial cells (stellate cells):
  - Located between basal lamina and epithelium
  - Contractile, regulate glomerular infiltration
  - Secrete extracellular matrix
  - Clear immune complexes
  - Involved in glomerular disease progression

- Filtration properties

- Free passage of neutral molecules is less than or equal to 4 nm
- Almost total exclusion of molecules is greater than or equal to 8 nm
- Filtration also depends on charge of molecules
- Total filtration area: greater than or equal to 0.8  $\text{m}^2$  per human kidney

- Tubular segments

- Proximal convoluted tubule (PCT)
  - Length: 15 mm; diameter: 55  $\mu\text{m}$
  - Single cell layers that interdigitate with one another and are united by apical tight junctions

- Brush border (microvilli) - increases surface area for absorption
- Lateral intercellular spaces - help in solute/water movement (extension of extracellular space)

- Loop of Henle

- Descending limb & thin ascending limb - thin, permeable cells
- Thick ascending limb - thick cells with many mitochondria (active transport)
- Cortical nephrons - short loops of Henle
- Juxtramedullary nephrons (15%) -> long loops deep into medulla -> important for urine concentration

- Juxtaglomerular apparatus (JGA)

- Formed by: macula densa (specialized distal tubule cells), lacis cells, granular cells (renin-secreting) in afferent arteriole
- Regulates renal blood flow & blood pressure

- Distal convoluted tubule (DCT)

- Length: 5 mm
- Lower epithelium than PCT, no brush border
- Fewer microvilli
- Begins at macula densa

- Collecting ducts

- Length: 20 mm
- Pass through cortex and medulla -> empty at renal pyramid apex
- Two main cell types:
  - Principle (P) cells - Na reabsorption, vasopressin-stimulated water reabsorption
  - Intercalated (I) cells - acid secretion,  $\text{HCO}_3^-$  transport
- Total nephron length (with ducts): 45-65 mm

- Secretory cells in the kidney

- Juxtaglomerular (granular) cells - secrete renin
- Renal medullary interstitial cells (RMICs):
  - Specialized fibroblast-like cells with lipid droplets
  - Major site of COX-2 and PGES - synthesize PGE2

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- PGE2: key paracrine regulator of salt & water balance
  - Other prostaglandins (e.g., PGE2) secreted by arterioles and glomeruli

## BLOOD VESSELS

### - Entry of blood

- Interlobar arteries -> give rise to afferent arterioles
- Each afferent arteriole have short, straight branch -> enters nephron

### - Glomerular capillaries

- Afferent arterioles divide into multiple branches -> forms glomerular tuft (capillaries)
- These capillaries then coalesce into efferent arterioles
- Special feature:
  - Only capillaries in the body that drain into arterioles (not venules)
  - Forms a portal system between glomerulus and tubules
- Smooth muscle:
  - Efferent arterioles have relatively little smooth muscle

### - Efferent arterioles

- After leaving the glomerulus, the efferent arterioles break up again into peritubular capillaries (supply nephron tubules)
- Important:
  - One afferent arteriole can supply multiple nephrons
  - Tubules don't always receive blood from "their own" glomerulus

### - Cortical vs juxtamedullary nephrons

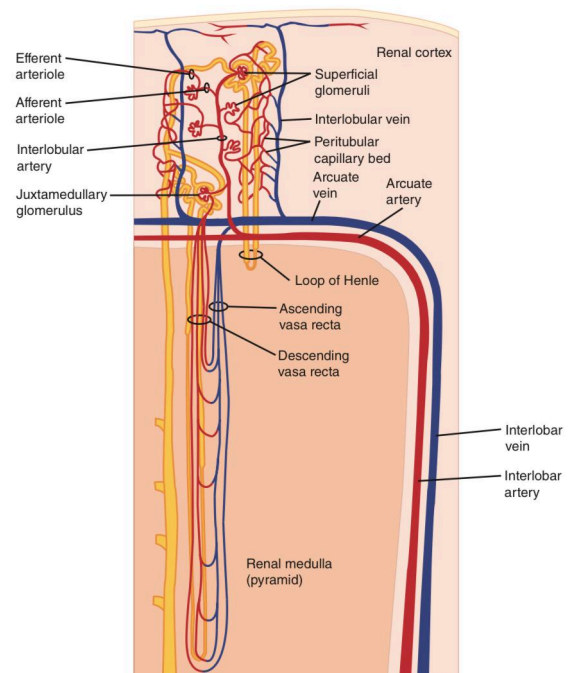
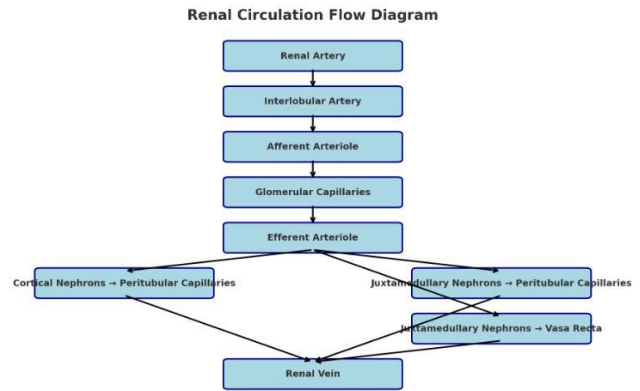
- Cortical nephron efferents -> only form peritubular network
- Juxtamedullary nephron efferents -> form:
  - Peritubular network and vasa recta (hairpin loops alongside loop of Henle)

### - Vasa recta

- Descending vasa recta:
  - Non fenestrated endothelium
  - Contain facilitated urea transporters
- Ascending vasa recta:
  - Fenestrated endothelium
  - Function: solute exchange - crucial for urine concentration

### - Capillary-tubule relationship

- Total renal capillary surface = total tubule surface area =  $12 \text{ m}^2$
- Blood volume in renal capillaries at any given time = 30-40 mL



## LYMPHATICS

- Kidneys drain via thoracic duct into the venous circulation in the thorax

## CAPSULE

- Thin but tough
- If kidney becomes edematous, the capsule limits the swelling, and the tissue pressure (renal interstitial pressure) rises

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- Decreases glomerular filtration rate (GFR) and as claimed to enhance and prolong anuria in acute kidney injury (AKI)

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## INNERVATION OF THE RENAL VESSELS

### - Pathway and components

- The renal nerves travel alongside the renal blood vessels as they enter the kidney
- They contain:
  - Postganglionic sympathetic efferent fibers (majority)
  - Afferent fibers (fewer)
  - Possible cholinergic fibers from the vagus nerve, though their roles remains uncertain

### - Sympathetic innervation

- Preganglionic fibers: originates from the lower thoracic and upper lumbar spinal cord segments
- Postganglionic neurons: located in the sympathetic chain ganglia, superior mesenteric ganglion, and along the renal artery
- Target sites of sympathetic fibers:
  - Afferent and efferent arterioles - regulate renal blood flow
  - Proximal and distant tubules - influence reabsorption/secretion
  - Juxtaglomerular apparatus - modulates renin release
  - Thick ascending limb of the loop of Henle - dense noradrenergic innervation (norepinephrine-mediated)

### - Afferent innervation

- Nociceptive afferents:
  - Parallel sympathetic efferents
  - Carry pain sensations (e.g., from kidney disease)
  - Enter the thoracic and upper lumbar dorsal roots
- Other renal afferents:
  - Participate in renorenal reflex:
    - If the ureteral pressure rises in one kidney, efferent sympathetic activity to the contralateral kidney decreases
    - This causes the opposite kidney to increase sodium and water excretion, balancing function

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## RENAL CIRCULATION

### BLOOD FLOW

#### - Renal blood flow in resting adult

- Kidney receive 1.2-1.3 L of blood per minute
- This equals to ~25% of cardiac output - shows the kidneys' high perfusion relative to size

#### - Measurement of renal blood flow - can be measured by:

- Electromagnetic or flow meters
- Fick principle - based on uptake/excretion of a substance relative to its arteriovenous concentration difference

#### - Renal plasma flow (RPF)

- Since the kidney filters plasma, not the whole blood, flow is expressed as RPF
- Formula:
  - $RPF = \frac{\text{amount of substance excreted per unit time}}{\text{arteriovenous difference of substance across kidney}}$
- Conditions for valid measurement:
  - Substance must not be metabolized, stored, or produced by the kidney
  - Must not alter renal blood flow itself

#### - Use of PAH (p-aminohippuric acid)

- PAH characteristics:
  - Freely filtered by glomeruli
  - Secreted by tubular cells
  - Very high extraction ratio (~90% removed in one pass through kidney at low dose)
- Because of this, PAH is used to estimate RPF

#### - Effective renal plasma flow (ERPF)

- Often, venous PAH is ignored in calculations -> gives effective RPF (ERPF)
- Formula:
  - $ERPF = \frac{\text{urinary PAH excretion rate}}{\text{plasma PAH concentration}}$
- Peripheral venous plasma can be used since its PAH = arterial PAH
- Normal ERPF in humans = 625 mL/min

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## PRESSURE IN RENAL VESSELS

### - Glomerular capillary pressure

- Direct measurements (in rats) show it is lower than predicted by indirect methods

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- At mean systemic arterial pressure of 100 mmHg, glomerular capillary pressure = 45 mmHg
- This equals about 40% of systemic arterial pressure (also seen in squirrel monkeys, and assumed similar in humans)
- Pressure drop across the glomerulus
  - The pressure difference between the afferent end and efferent and of the glomerulus is very small -> only 1-3 mmHg
  - Indicated low resistance inside the glomerular capillaries
- Efferent arteriole and peritubular capillaries
  - A larger pressure drop occurs in the efferent arteriole
  - As blood enters peritubular capillaries, pressure falls to about 8 mmHg
- Renal venous pressure
  - Blood then drains into the renal vein, where pressure is about 4 mmHg

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## REGULATION OF RENAL BLOOD FLOW

### Vasoconstrictors

#### 1.) Norepinephrine (noradrenaline)

- Constrict renal vessels, strongest effect on:
  - Interlobar arteries and afferent arterioles
- Low renal blood flow decrease GFR (if excessive)

#### 2.) Angiotensin II

- Constricts both afferent and efferent arterioles
- Helps maintain filtration pressure when systemic BP falls

### Vasodilators

#### 1.) Dopamine (produced by kidney)

- Causes renal vasodilation
- Promotes natriuresis (excretion of sodium)

#### 2.) Prostaglandins (PGs)

- Increase cortical blood flow
- Decrease medullary blood flow (protects against hypoxia)

#### 3.) Acetylcholine

- Produces renal vasodilation

- Dietary influence

- High protein diet
  - Raises glomerular capillary pressure
  - Increases renal blood flow

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## FUNCTIONS OF THE RENAL NERVE

### 1.) Renin secretion

#### - Mechanism

- Norepinephrine released from renal nerves binds to B1-adrenergic receptors on juxtaglomerular (JG) cells
  - Increases renin secretion

#### - Order of response (with increasing stimulation):

- Increase sensitivity of granular (JG) cells
- Increase renin secretion
- Increase Na reabsorption
- At highest threshold -> renal vasoconstriction - decreased GFR & renal blood flow (RBF)

### 2.) Sodium reabsorption

#### - Direct effect of norepinephrine on renal tubular cells

#### - Occurs in:

- Proximal tubule
- Distal tubule
- Thick ascending limb of Henle's loop

#### - Mediated by $\alpha$ -adrenergic and/or B-adrenergic receptors (exact mechanism unsettled)

### 3.) Renal vasoconstriction

#### - Strong sympathetic stimulation -> marked decrease in RBF

#### - Receptor mechanisms:

- $\alpha$ 1-adrenergic receptors (major)
- $\alpha$ 2-adrenergic receptors (minor)

### 4.) Tonic activity & reflex control

#### - Renal nerves have baseline tonic discharge at rest

#### - Systemic BP drop:

- Decrease baroreceptor nerve activity -> reflex renal vasoconstriction

#### - During exercise & standing up (from supine position):

- Renal blood flow decreases (helps redistribute blood to muscles & maintain systemic BP)

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## AUTOREGULATION OF RENAL BLOOD FLOW

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- Ability of the kidney to maintain relatively constant RBF despite changes in perfusion pressure (90-220 mmHg in dogs)

- Ensures stable glomerular filtration (GFR) and kidney function

- Key features:

- Present in: denervated kidneys and isolated perfused kidneys
- Lost when vascular smooth muscle is paralyzed  
-> confirms dependence on intrinsic vascular mechanism (not nerves)

- Mechanism involved

- Myogenic response (afferent arteriole)
  - Increase perfusion pressure - stretches afferent arteriole -> direct contraction of vascular smooth muscle
  - Prevents excess increase in RBF and GFR
- Role of NO
  - May contribute by modulating vasodilation to balance constriction
- Angiotensin II (at low pressures)
  - Constrict efferent arteriole - help maintain GFR when renal perfusion pressure is low
  - Clinical relevance: if ACE inhibitors are given to patients with already poor renal perfusion, angiotensin II production is blocked -> efferent arteriole dilates -> drop in GFR -> renal failure

- Clinical relevance:

- ACE inhibitors/ARBs can worsen kidney function in patients with renal artery stenosis or hypotension, because they prevent angiotensin II-mediated efferent constriction

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## REGIONAL BLOOD FLOW & O<sub>2</sub> CONSUMPTION

1.) Renal cortex

- Function: filtration of large blood volumes through glomeruli

- Blood flow: very high (~5 mL/g/min) - higher than the brain (0.5 mL/g/min)

- O<sub>2</sub> extraction: relatively low

- Arteriovenous O<sub>2</sub> difference = 14 mL/L (vs. brain 62 mL/L; heart 114 mL/L)

- Cortical PO<sub>2</sub>: ~50mmHg

- Clinical relevance: cortex gets a lot of blood for filtration, so O<sub>2</sub> supply exceeds its metabolic demand

2.) Renal medulla

- Function: maintaining the osmotic gradient necessary for urine concentration

- Blood flow: much higher than cortex

- Outer medulla: ~ 2.5 mL/g/min
- Inner medulla: ~ 0.6 mL/g/min

- O<sub>2</sub> extraction: relatively high because of active Na reabsorption in the thick ascending limb of Henle

- Medullary PO<sub>2</sub>: ~ 15 mmHg (very low)

- Clinical relevance: medulla is vulnerable to hypoxia if blood flow is further reduced (ischemia risk)

3.) Local regulation

- To maintain balance between low blood flow and high metabolic needs, local factors act in a paracrine manner:

- NO, prostaglandins, and cardiovascular peptides

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## GLOMERULAR FILTRATION

### MEASURING GLOMERULAR FILTRATION RATE

- GFR - the amount of plasma ultrafiltrate formed per minute in the glomeruli

- Unit: mL/min

- Clinical importance: it is a key measure of kidney function

- Normal GFR: 123 mL/min

- Requirements for substance to measure GFR:

- Freely filtered at the glomerulus
- Not reabsorbed by tubules
- Not secreted by tubules
- Not metabolized in the body
- Nontoxic

- Example: Inulin

- Inulin = polymer of fructose
- Meets all criteria -> gold standard
- Procedure:
  - Give IV loading dose of inulin
  - Maintain constant plasma level with sustaining infusion
  - Collect times urine sample and a plasma sample (midpoint)

- Renal plasma clearance (CX)

- Volume of plasma from which a substance is completely removed by the kidney per minute

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- Formula:

$$C_X = \frac{U_X \times V}{P_X}$$

Where:

- $U_X$  = concentration of substance X in urine
- $V$  = urine flow rate (mL/min)
- $P_X$  = concentration of substance X in plasma
- If the substance behaves like inulin → Clearance = GFR.

| Substance  | Clearance (mL/min) |
|------------|--------------------|
| Glucose    | 0                  |
| Sodium     | 0.9                |
| Chloride   | 1.3                |
| Potassium  | 12                 |
| Phosphate  | 25                 |
| Urea       | 75                 |
| Inulin     | 125                |
| Creatinine | 140                |
| PAH        | 560                |

## PERMEABILITY

### 1.) High permeability

- Glomerular capillaries are about 50x more permeable than skeletal muscle capillaries
- They allow passage of water and small solutes but restrict proteins and larger molecules

### 2.) Size selectivity

- < 4 nm (neutral molecules): freely filtered
- > 8 nm: filtration = 0
- 4-8 nm: filtration decreases inversely with size

### 3.) Charge selectivity

- The glomerular basement membrane (GBM) has negatively charged sialoproteins

- Effect:

- Anions: filtration reduced (repelled)
- Cations: filtration enhanced (attracted)
- Neutrals: baseline permeability, dependent only on size

- Example: albumin

- Effective molecular diameter = 7 nm, negatively charged

- Based on size alone -> should be filtered more than observed
- In reality -> glomerular concentration is only 0.2% of plasma concentration

- Explanation: charge barrier repels albumin

- Example: protein in urine

- Normal: < 100 mg/day
  - Mostly from shed tubular cells, not filtration
- Albuminuria: presence of significant albumin in urine
  - Can occur in nephritis when negative charges in GBM are lost, even without an increase in pore size

## SIZE OF CAPILLARY BEDS

- Kf (filtration coefficient) - measure of how easily fluid passes through the glomerular capillaries

- Determined by:

- Permeability of the capillary wall
- Surface area available for filtration

- Role of meningeal cells

- Location: found between glomerular capillaries
- Function: contractile cells that regulates surface area for filtration
- When meningeal cell contract:
  - Less Kf - less filtration
    - Capillary surface is reduced
    - Contraction at bifurcation points shifts blood flow away from certain loops
    - Contraction also distorts and narrows capillary lumens

- Regulators of mesangial cell contraction

- Various agents influence mesangial cells

| Contraction                                    | Relaxation       |
|------------------------------------------------|------------------|
| Endothelins                                    | ANP              |
| Angiotensin II                                 | Dopamine         |
| Vasopressin                                    | PGE <sub>2</sub> |
| Norepinephrine                                 | cAMP             |
| Platelet-activating factor                     |                  |
| Platelet-derived growth factor                 |                  |
| Thromboxane A <sub>2</sub>                     |                  |
| PGF <sub>2</sub>                               |                  |
| Leukotrienes C <sub>4</sub> and D <sub>4</sub> |                  |
| Histamine                                      |                  |

- Key example:

- Angiotensin II: strong stimulator of mesangial contraction

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- Glomeruli contain Angiotensin II receptors
- Additional note:
  - Mesangial cells may also produce renin, contributing to local renin-angiotensin system activity

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## HYDROSTATIC & OSMOTIC PRESSURE

- Glomerular capillaries have higher pressure than other capillaries

- Afferent arterioles: short, straight branches from interlobar arteries -> allow high pressure transmission into the glomerulus
- Efferent arterioles: provide relatively high resistance downstream -> this maintains a higher pressure inside the glomerular capillaries compared to systemic capillaries

- Since PGC is uniquely high due to afferent input and efferent resistance, the kidney maintains strong filtration pressure

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## CHANGES IN GFR

- Renal vascular resistance as a result of autoregulation tend to stabilize filtration pressure

- Efferent constriction > afferent constriction

- Mean systemic arterial pressure drops below the autoregulatory range, GFR drops

- Either type of constriction decreases blood flow to the tubules

- Filtration fraction

- Ratio between GFR to the RPF
- Normally between 0.16 to 0.20
- Fall of systemic blood pressure -> GFR falls less than RPF because of efferent arteriole constriction, and consequently the filtration fraction rises

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| Changes in renal blood flow                                                                    |
| Changes in glomerular capillary hydrostatic pressure                                           |
| Changes in systemic blood pressure                                                             |
| Afferent or efferent arteriolar constriction                                                   |
| Changes in hydrostatic pressure in Bowman capsule                                              |
| Ureteral obstruction                                                                           |
| Edema of kidney inside tight renal capsule                                                     |
| Changes in concentration of plasma proteins: dehydration, hypoproteinemia, etc (minor factors) |
| Changes in $K_f$                                                                               |
| Changes in glomerular capillary permeability                                                   |
| Changes in effective filtration surface area                                                   |

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## TUBULAR FUNCTION

### MECHANISM OF TUBULAR REABSORPTION & SECRETION

- Reabsorption of proteins and peptides

- Small proteins & peptide hormones - reabsorbed in the proximal tubule via endocytosis
- Once inside, they are degraded to amino acids, which return to circulation

- Pathways of solute transport

- Transport in the renal tubules occurs by:
  - Passive diffusion
    - Paracellular or transcellular
    - Follows chemical or electrical gradients
  - Facilitated diffusion
    - Via specific carrier proteins down gradients
  - Active transport
    - Moves solutes against gradients using ATP or ion-coupled transport
- Transport is mediated by:
  - Ion channels, exchangers (antiporters), cotransporters (symporters), pumps

- Membrane polarity

- Luminal (apical) membrane transporters is not equal to basolateral membrane transporters
- This polarized distribution allow directional, net transport of solutes from tubular fluid -> blood
  - E.g., Na enters cell via apical cotransporter, exits via basolateral Na/K ATPase

- Transport maximum

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- Each active transport system has a transport maximum ( $T_m$ )
  - Below  $T_m$  - transport increases with concentration
  - At/above  $T_m$  - carriers are saturated, no further increase in transport rate
- Some transporters (e.g., Na-glucose cotransporters) have very high  $T_m$ , so saturation is rate physiologically
- Epithelial “leakiness” and paracellular pathway
  - Tubular epithelium (especially proximal tubule) is leaky
    - Tight junctions allow passage of water and electrolytes
    - This paracellular pathway contributes significantly to overall transport
  - Evidence: Paracellin-1 (tight junction protein) is crucial for Mg reabsorption
    - Loss-of-function - severe Mg and Ca loss of urine

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## NA REABSORPTION

- General mechanism
  - Entry into tubular cells (apical side)
    - By cotransport (symport) or exchange (antiport) depending on the nephron segment
  - Exit into interstitium (basolateral side)
    - By Na/K ATPase pump (universal to all nephron segments except thin loop of Henle)
- Chloride handling
  - Proximal tubule: Cl reabsorbed paracellularly and via cotransport with Na (later segments)
  - TAL: major Cl reabsorption with Na and K via NKCC2 cotransporter
  - Distal tubule: with Na via NCC cotransporter
  - Collecting duct: follows Na either via paracellular route or coupled transport
- Paracellular pathway
  - Na and Cl can also move between cells via tight junctions especially in proximal tubule and thick ascending limb
  - This contributes significantly to overall Na reabsorption
- Hormonal regulation
  - Aldosterone (distal nephron & collecting duct):

- Increase ENaC expression -> increase Na reabsorption
- ANP/BNP (atrial/brain natriuretic peptides):
  - Inhibit Na reabsorption in collecting ducts

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## GLUCOSE REABSORPTION

- Site & mechanism
  - Location: early proximal tubule
  - Mechanism: secondary active transport
    - Apical side (tubular lumen -> cell): Na-glucose cotransporters (SGLTs)
      - SGLT2 (proximal S1/S2): high-capacity, low-affinity -> reabsorbs ~90%
      - SGLT1 (proximal S3): low-capacity, high-affinity -> reabsorbs ~10%
    - Basolateral side (cell -> blood): facilitated diffusion via GLUT2/GLUT1 transporters
  - Couples to Na movement down its electrochemical gradient (maintained by Na/K ATPase)
- Normal handling
  - Filtered glucose load:
    - $PG \sim 80 \text{ mg/dL} \times GFR \text{ } 125 \text{ mL/min} \approx 100 \text{ mg/min}$
  - Reabsorption: ~100% under normal plasma glucose (PG) levels
  - Urine glucose: only trace amounts (< 25 mg/day)
- Transport maximum ( $T_mG$ )
  - Maximal capacity of glucose transporters
  - $T_mG$  values:
    - Men: ~375 mg/min
    - Woman: ~300 mg/min
  - Beyond  $T_mG$  -> transporters saturated -> glucose appears in urine (glucosuria)
- Renal threshold
  - Predicted threshold:  $375 \text{ mg/min} / 125 \text{ mL/min} = \sim 300 \text{ mg/dL PG}$
  - Actual glucose: ~200 mg/dL arterial plasma (~180 mg/dL venous)
  - Lower than predicted because:
    - Not all nephrons have identical  $T_m$
    - Some glucose “escapes” reabsorption before saturation

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## - Splay

- The rounded portion of the glucose reabsorption curve via threshold
- Reflects:
  - Heterogeneity of nephron transporters
  - Variability in binding avidity of SGLTs
- Means glucose begins to appear in urine gradually, not abruptly, as PG arises

## - Clinical notes

- Glucosuria causes:
  - Diabetes mellitus (PG exceeds threshold)
  - Pregnancy (increased GFR → filtered load exceeds  $T_m$ )
  - Renal glycosuria (rare genetic defects in SGLTs)
- SGLT2 inhibitors (e.g., dapagliflozin, empagliflozin)
  - Block glucose reabsorption in proximal tubule → promote glucosuria → lower blood glucose

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## TUBOGLOMERULAR FEEDBACK & GLOMERUTUBULAR BALANCE

### - Tuboglomerular feedback

- A feedback mechanism in which the renal tubule regulates its own glomerulus to stabilize GFR
- Maintains a relatively constant delivery of fluid/solute to the distal nephron

### - How it works:

- Sensor: macula densa (distal tubule, near glomerulus)
  - Specialized cells detect Na & Cl concentration in tubular fluid
- Signal transduction:
  - Na & Cl enter macula densa via Na-K-2Cl cotransporter (NKCC2)
  - NaCl increase also increase Na/K ATPase activity, which leads to increase ATP hydrolysis
  - ATP breakdown increases adenosine production
- Effector response:
  - Adenosine is released basolaterally
  - Acts on A1 receptors on afferent arteriole smooth muscle

- Causes increase of intracellular Ca, which causes vasoconstriction and lowers GFR

### • Opposite scenario:

- If NaCl delivery to macula densa is low, adenosine release decreases
- Afferent arteriole dilates → increase GFR

### - Functional role:

- Prevents excessive Na and water loss when flow is high
- Prevents under-filtration when flow is low
- May also signal to juxtaglomerular cells → decreases renin release (though this part is less certain)

### - Glomerulotubular balance (GTB)

- The ability of the proximal tubule to reabsorb a constant fraction (~65%) of the filtered Na and water, even if GFR changes

### - Mechanism:

- When GFR increases, more Na and H<sub>2</sub>O are filtered
- Peritubular capillary oncotic pressure increases (because more plasma protein is concentrated as fluid leaves via efferent arteriole)
- Higher oncotic pressure enhances proximal tubular reabsorption of Na and H<sub>2</sub>O
- Maintains balance: filtration and reabsorption rise together → prevents overload of distal nephron

### - Functional role:

- Ensures proportional reabsorption of solutes and H<sub>2</sub>O
- Protects against massive fluid/electrolyte shifts when GFR changes

| Feature  | Tubuloglomerular Feedback (TGF)                          | Glomerulotubular Balance (GTB)             |
|----------|----------------------------------------------------------|--------------------------------------------|
| Location | Macula densa → afferent arteriole                        | Proximal tubule & peritubular capillaries  |
| Trigger  | NaCl delivery to distal tubule                           | Changes in GFR & oncotic pressure          |
| Response | Adjusts GFR via afferent arteriole constriction/dilation | Adjusts reabsorption to match filtration   |
| Purpose  | Stabilizes GFR and distal delivery                       | Maintains constant fractional reabsorption |
| Speed    | Very fast (seconds)                                      | Fast but slightly slower (seconds–minutes) |

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## AQUAPORINS “specialized water channels”

- integral membrane proteins that allow rapid diffusion of water across cell membrane
- They make water transport fast, selective, and regulated, compared to slow passive diffusion through bilipid layers
- 13 aquaporins have been cloned in humans
  - Crucial: AQP 1-4
- Roles of aquaporins in the kidney
  - AQP1
    - Location: proximal tubule + descending thin limb of Henle's loop
    - Function: responsible for bulk water reabsorption ( $\approx 70\%$  of filtered water)
    - Control: constitutively expressed (not regulated by hormones)
  - AQP2
    - Location: apical (luminal) and basolateral membranes of proximal tubules epithelial cells
    - Function: regulated water reabsorption in response to vasopressin (ADH); provide high water permeability
    - Mechanism:
      - ADH binds to V2 receptors -> increases CAMP -> activates PKA
      - PKA phosphorylates AQP2-containing vesicles -> vesicles fuse with apical membrane
      - This inserts AQP2 channels into the luminal surface -> water enters the cell
    - Pathology: Lack of AQP2 response -> nephrogenic diabetes insipidus (kidneys cannot concentrate urine)
  - AQP3 & 4
    - Location: basolateral membrane of collecting duct principal cells
    - Function: allow water to exit cells into interstitium after entering via AQP2
    - Control: not directly regulated by ADH, but essential for water movement
- Overall mechanism in collecting duct:

1. ADH  $\rightarrow$  inserts **AQP2** in apical membrane.
2. Water moves from **tubule lumen**  $\rightarrow$  **epithelial cell** via AQP2.
3. Water exits **epithelial cell**  $\rightarrow$  **interstitium** via **AQP3 & AQP4**.
4. Result: Water reabsorption increases  $\rightarrow$  urine becomes concentrated.

## PROXIMAL TUBULE

- Active transport + water flow in the proximal tubule
  - Many solutes (Na, glucose, amino acids, etc.) are actively reabsorbed in the proximal tubules
  - Despite solute transport, the fluid in the lumen remains iso-osmotic with plasma all the way to the end of the proximal tubules
  - This is possible because AQP-1 is present in both apical and basolateral membrane, allowing rapid water movement along osmotic gradients
  - Result: water exits as solutes exit, so isotonicity is maintained
- Micropuncture studies (TF/P ratio)
  - Inulin is used as a marker (not reabsorbed)
  - At the end of the proximal tubule, TF/P inulin ratio = 2.5 to 3.3
  - This indicates that 60-70% of filtered water and solutes have already been reabsorbed by this point
- Human mutations in AQP1
  - People lacking AQP1 function do not have as severe defect, however still have an impaired ability to respond to dehydration

## LOOP OF HENLE

- Juxtamedullary nephrons & medullary gradient
  - Long loops of Henle dip deep into the medulla
  - The medullary interstitium becomes progressively hyperosmotic (up to  $\sim 1200$  mOsm/kg near the papilla)
  - This gradient is essential for water reabsorption from collecting ducts under vasopressin
- Descending limb
  - Permeable to water (AQP1 channels on both apical and basolateral membranes)
  - Not permeable to solutes -> water exits into hypertonic medullary interstitium, making tubular fluid hypertonic
- Ascending limb
  - Impermeable to water

# WEEK 1: INTEGRATIVE PHYSIOLOGY

- Has Na-K-2Cl cotransporter (NKCC2) in the thick ascending limb
- Na is actively pumped into interstitium (via Na/K ATPase), Cl diffuses out through ClC-Kb channels, K recycles via ROMK channels
- Result: tubular fluid becomes hypotonic as solutes leave but water cannot flow
- This segment is called the diluting segment
- Water balance
  - About 15% of filtered water is reabsorbed in the descending loop of Henle
  - By the time the fluid reaches the distal tubule, ~20% of filtered water remains
  - TP/P inulin ratio  $\approx 5$ , meaning the filtrate is concentrated compared to plasma
- Transporter
  - NKCC2 is a secondary active transporter (driven by Na gradient maintained by Na/K ATPase)
  - Found also in GI tract, airways, and salivary glands

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## DISTAL TUBULE

- Relatively impermeable to water, and continued removal of the solute in excess of solvent further dilutes the tubular fluid

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## COLLECTING DUCTS

- Structure
  - Two parts:
    - Cortical portion (in cortex)
    - Medullary portion (in medulla, ends in renal pelvis)
  - Function depends on vasopressin (ADH) from the posterior pituitary
- Role of vasopressin:
  - ADH increase water permeability of the collecting ducts
  - Key player: AQP2
    - Stored in vesicles inside principal cells
    - Upon ADH binding to V2 receptor, a cAMP-Protein Kinase A (PKA) pathway is activated
    - Triggers insertion of AQP2 into the apical membrane -> water can now move out

- Cytoskeletal proteins (dynein, dynactin, myosin-1) help shuttle AQP2 to the membrane
- Collecting ducts with vasopressin (maximal antidiuresis)
  - Cortical collecting ducts
    - Hypotonic fluid enters
    - Water leaves -> fluid becomes isotonic
    - Up to 10% of filtered water is reabsorbed here
  - Medullary collecting ducts
    - Isotonic fluid enters with TF/P inulin  $\approx 20$
    - Additional 4.7% water is reabsorbed into the hypotonic medullary interstitium
    - Urine becomes highly concentrated (TF/P > 300)
  - Maximum urine osmolality in humans:  $\sim 1400$  mOsm/kg ( $\approx 5\times$  plasma)
  - Total water reabsorbed: up to 99.7% of filtered water
- Collecting ducts w/o vasopressin
  - Collecting ducts epithelium = relatively impermeable to water
  - Tubular fluid stays hypotonic -> large volume of dilute urine flows to the pelvis
  - Urine osmolality can be as low as 30 mOsm/kg
  - Even without ADH:
    - 2% of filtered water is still reabsorbed
    - 13% of filtered water may be excreted
    - Urine flow can reach 15 mL/min

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## ROLE OF UREA

- Helps establish the osmotic gradient in the medullary pyramids
- This gradient is essential for the kidney's ability to produce concentrated urine
- Urea transport mechanism
  - Urea transport occurs via urea transporters (UTs) through facilitated diffusion
    - UT-A1 - collecting ducts
    - UT-A2 - loop of Henle
    - UT-A3 - collecting ducts
    - UT-A4 - inner medulla
    - UT-B - erythrocytes & descending vasa recta
- Urea transport in the collecting ducts
  - Mediated mainly by UT-A1 and UT-A3
  - Both are regulated by vasopressin

# WEEK 1: INTEGRATIVE PHYSIOLOGY

- High vasopressin (antidiuresis):
  - Increase urea reabsorption into medullary interstitium
  - Strengthens medullary osmotic gradient
  - Increase concentrating ability of the kidney
- Effects of protein intake on urea and urine concentration
  - Urea in the medullary interstitium (and ultimately urine) depends on how much is filtered
  - Urea production is directly related to dietary protein intake
    - High-protein diet: increase urea filtered -> stronger osmotic gradient -> better urine concentration
    - Low-protein diet: decrease urea filtered -> weaker osmotic gradient -> reduced urine concentration

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## REGULATION OF WATER EXCRETION

### WATER DIURESIS

- Control of vasopressin secretion
  - Regulated by plasma osmotic pressure
    - Increased plasma osmolality - stimulates vasopressin release
    - Decreased plasma osmolality - inhibits vasopressin release
- Effect of drinking hypotonic fluid (water load)
  - Drinking large amounts of water (low osmolality) leads to water diuresis
  - Timeline:
    - Starts ~15 minutes after ingestion
    - Peaks at ~40 minutes
- Early response (before water absorption)
  - The act of drinking itself produces a small, immediate decrease in vasopressin secretion
  - This occurs before the water is absorbed into the bloodstream
- Main response (after water absorption)
  - Most of the inhibition of vasopressin secretion comes from the fall in plasma osmolality once the ingested water is absorbed
  - Result: reduced water reabsorption in the kidneys -> increased urine output (diuresis)
- Maximal urine flow during water diuresis
  - The kidneys can excrete up to ~16 mL/min of urine when clearing an average osmotic load

- If water is ingested faster than this, the body cannot excrete it quickly enough
- Result of excess water intake
  - ECF becomes hypotonic
- Water intoxication
  - Rate but dangerous condition caused by excessive water intake
  - Brain cells are vulnerable
    - Swelling -> convulsion, coma, or death
  - Other causes:
    - Exogenous vasopressin (if water intake isn't restricted)
    - Endogenous vasopressin secretion triggered by nonosmotic stimuli (e.g., surgical trauma)
    - Oxytocin administration after birth if water intake is not carefully monitored

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## THE BLADDER

### FILLING

- Peristalsis
  - Regular peristaltic contractions (1-5 /min)
  - Function: propel urine from the renal pelvis to bladder
  - Urine enters the bladder in spurts, each corresponding to a peristaltic wave
- Bladder entry
  - Ureters enter the bladder obliquely through its wall
  - No true ureteral sphincters exist, but:
    - The obliquely pathway compresses ureters when the bladder fills
    - This prevents backflow (reflux) of urine into the ureters

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### EMPTYING

- Detrusor muscle: circular muscle layer -> main muscle responsible for bladder emptying
- Sphincters
  - Internal urethral sphincter
    - Not a true sphincter; consists of smooth muscle bundles around the urethra
    - Function (especially in males): prevent some reflux into the bladder during ejaculation
  - External urethral sphincter

# WEEK 1: INTEGRATIVE PHYSIOLOGY

- Made of skeletal muscle
- Located in the membranous urethra
- Function: allows conscious control of urination
- Bladder epithelium
  - Superficial layer: flat cells
  - Deep layer: cuboidal cells (specialized transitional epithelium -> stretchy)
- Bladder filling & pressure (cystometry)
  - Filling behavior: urine enters with little rise in pressure until bladder is fairly full
  - Plasticity of smooth muscle: when stretched, tension initially rises but then relaxes
    - Urge to void: felt around 150 mL
  - Strong sense of fullness: around 400 mL
- Micturition (urination process)
  - Reflex nature: a spinal reflex, modulated by higher brain centers
  - Steps during urination:
    - Relaxation of pelvic floor muscles & external sphincter
    - Detrusor muscle contracts -> bladder pressure rises
    - Urine flows through urethra
  - Voluntary control: contraction of external sphincter & perineal muscles can delay or stop urination
- After urination:
  - Females: urethra empties by gravity
  - Males: remaining urine by bulbocavernosus muscle contractions

## FUNCTION OF KIDNEYS

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## PART OF KIDNEY

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## EXTERNAL

- adrenal glands: none, epi, glucocorticoids
- retroperitoneal: outside the peritoneum: covered by adipose tissue for protection

## RENAL BLOOD SUPPLY

- glomerular (from glomerulus = around 60 mmHg) and peritubular (filtered blood = 13 mmHg) capillaries

## NEPHRON

- cortical (found in the renal cortex) - loop of Henle
- juxtaglomerular
- Bowman's - proximal tubule - descending limb - thin segment of ascending limb - thick segment - loop of Henle

- Major parts:

- Renal corpuscle: BV, Bowman's capsule, JG cell (renin)
- Renal tubules

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## ENDOCRINE FUNCTION

- Erythropoiesis -> Increases EPO -> increased oxygen in blood
- Calciferol (Vitamin D) -> increased absorption of Calcium
- Prostaglandin - vasodilation
- Dopamine - formed in PCT cell - natriuresis (urination)
  - Pressors for near death patient

## URINE FORMATION

### - Glomerular filtration

- Glomerular capillaries - exchange
- Filter out water, solutes
- Glomerular filtration rate: 125 mL/min in male; 105 mL/min in female
- Afferent arteriole, glomerular capillaries, Bowman's capsule (filtration), efferent arteriole

### - Tubular reabsorption

- For water (65%) and solutes
- Reabsorption in PCT
- Concentrated urine because of reabsorption
- First is reabsorption in water in PCT and then solutes, in DCT slightly diluted urine
- Medullary collecting duct
  - Less reabsorption because of PCT
  - final processing of urine and final urine output

### - Secretion

- Excretion = amount filtered - amount reabsorbed + amount secreted
- < 50 mL = normal urine volume of the bladder after urination
- 500 mL = stored urine

## GLOMERULAR FILTRATION RATE

- Normal: 125 mL/day or 180 L/day

# WEEK 1: INTEGRATIVE PHYSIOLOGY

- 1 mL /kg/hour = 1,200 mL/day
- Higher GFR in male than female
- Decrease in filtration = increase in creatinine

## STARTING FORCES

- Hydrostatic - pushes out solutes
- Oncotic - pull in solutes from interstitium to the capillaries

## TABLE

- Aldosterone - hypokalemia

UTI = pagpigil ng ihi = urine is solute concentrated =  
microbacteria accumulates